

CS166 Lab 9 Assignment

Optimization

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Query Plan 1 (Hash Join)

- In the first query plan, the optimizer chooses a hash join between sales and catalog with a nested hash join to stores, and performs sequential scans on sales, stores, and catalog. This plan doesn't benefit from the new index and relies on hashing the join inputs.

```
ERROR: could not open file "stores.csv" for reading: No such file or directory
HINT: COPY FROM instructs the PostgreSQL server process to read a file. You may want a client-side facility such as psql's \copy.
ERROR: could not open file "sales.csv" for reading: No such file or directory
HINT: COPY FROM instructs the PostgreSQL server process to read a file. You may want a client-side facility such as psql's \copy.
ERROR: could not open file "catalog.csv" for reading: No such file or directory
HINT: COPY FROM instructs the PostgreSQL server process to read a file. You may want a client-side facility such as psql's \copy.
[dtope004@xe-10 lab9_updated]$ cs166_psql $USER'_lab9_DB' < query.sql
QUERY PLAN
-----
Hash Join (cost=33.72..69.77 rows=1700 width=492) (actual time=0.019..0.020 rows=0 loops=1)
  Hash Cond: (sales.itemid = catalog.itemid)
  -> Hash Join (cost=16.52..48.05 rows=1700 width=270) (never executed)
    Hash Cond: (sales.storeid = stores.storeid)
    -> Seq Scan on sales (cost=0.00..27.00 rows=1700 width=20) (never executed)
    -> Hash (cost=12.90..12.90 rows=290 width=250) (never executed)
      -> Seq Scan on stores (cost=0.00..12.90 rows=290 width=250) (never executed)
  -> Hash (cost=13.20..13.20 rows=320 width=222) (actual time=0.002..0.002 rows=0 loops=1)
    Buckets: 1024 Batches: 1 Memory Usage: 8kB
    -> Seq Scan on catalog (cost=0.00..13.20 rows=320 width=222) (actual time=0.002..0.002 rows=0 loops=1)
Planning time: 0.638 ms
Execution time: 0.062 ms
(12 rows)

QUERY PLAN
-----
Nested Loop (cost=4.36..29.22 rows=6 width=492) (actual time=0.001..0.002 rows=0 loops=1)
 -> Nested Loop (cost=4.21..28.08 rows=6 width=270) (actual time=0.001..0.001 rows=0 loops=1)
   -> Seq Scan on stores (cost=0.00..13.62 rows=1 width=250) (actual time=0.001..0.001 rows=0 loops=1)
     Filter: ((state)::text = 'CA'::text)
   -> Bitmap Heap Scan on sales (cost=4.21..14.37 rows=8 width=20) (never executed)
     Recheck Cond: (storeid = stores.storeid)
     -> Bitmap Index Scan on sales_pkey (cost=0.00..4.21 rows=8 width=0) (never executed)
       Index Cond: (storeid = stores.storeid)
   -> Index Scan using catalog_pkey on catalog (cost=0.15..0.19 rows=1 width=222) (never executed)
     Index Cond: (itemid = sales.itemid)
Planning time: 0.160 ms
Execution time: 0.031 ms
(12 rows)

[dtope004@xe-10 lab9_updated]$ cs166_psql $USER'_lab9_DB' < index.sql
```

Query Plan 2 (Nested Loop Join)

- In the second query plan, the optimizer uses a nested loop join and after creating the index it switches the stores scan to an Index Scan using stateindex, showing that the index improves the state filter lookup while catalog still uses its primary key index for the join.

```
[dtope004@xe-10 lab9_updated]$ cs166_psql $USER'_lab9_DB' < index.sql
CREATE INDEX
CREATE INDEX
CREATE INDEX
[dtope004@xe-10 lab9_updated]$ cs166_psql $USER'_lab9_DB' < query.sql
QUERY PLAN
-----
Hash Join (cost=33.72..69.77 rows=1700 width=492) (actual time=0.019..0.020 rows=0 loops=1)
  Hash Cond: (sales.itemid = catalog.itemid)
  -> Hash Join (cost=16.52..48.05 rows=1700 width=270) (never executed)
    Hash Cond: (sales.storeid = stores.storeid)
    -> Seq Scan on sales (cost=0.00..27.00 rows=1700 width=20) (never executed)
    -> Hash (cost=12.90..12.90 rows=290 width=250) (never executed)
      -> Seq Scan on stores (cost=0.00..12.90 rows=290 width=250) (never executed)
  -> Hash (cost=13.20..13.20 rows=320 width=222) (actual time=0.002..0.002 rows=0 loops=1)
    Buckets: 1024 Batches: 1 Memory Usage: 8kB
    -> Seq Scan on catalog (cost=0.00..13.20 rows=320 width=222) (actual time=0.001..0.001 rows=0 loops=1)
Planning time: 0.597 ms
Execution time: 0.060 ms
(12 rows)

QUERY PLAN
-----
Nested Loop (cost=4.51..23.76 rows=6 width=492) (actual time=0.009..0.009 rows=0 loops=1)
 -> Nested Loop (cost=4.36..22.62 rows=6 width=270) (actual time=0.008..0.009 rows=0 loops=1)
   -> Index Scan using stateindex on stores (cost=0.15..8.17 rows=1 width=250) (actual time=0.008..0.008 rows=0 loops=1)
     Index Cond: ((state)::text = 'CA'::text)
   -> Bitmap Heap Scan on sales (cost=4.21..14.37 rows=8 width=20) (never executed)
     Recheck Cond: (storeid = stores.storeid)
     -> Bitmap Index Scan on sales_pkey (cost=0.00..4.21 rows=8 width=0) (never executed)
       Index Cond: (storeid = stores.storeid)
   -> Index Scan using catalog_pkey on catalog (cost=0.15..0.19 rows=1 width=222) (never executed)
     Index Cond: (itemid = sales.itemid)
Planning time: 0.145 ms
Execution time: 0.039 ms
(12 rows)
```